

Delinquency Prediction Dataset – EDA Summary Report

1. Dataset Overview

The dataset contains **500 customer records** and **19 variables** related to customer demographics, financial behavior, repayment history, and delinquency status.

Key Variables

- Demographics: Age, Location, Employment_Status
- Financial indicators: Income, Credit_Score, Loan_Balance, Debt_to_Income_Ratio, Credit_Utilization
- Behavioral indicators: Missed_Payments, monthly repayment status (Month_1 to Month_6)
- Target variable: Delinquent_Account

Dataset Characteristics

- Delinquent accounts represent approximately **16%** of the dataset.
- Non-delinquent accounts represent approximately **84%**.
- This indicates a **class imbalance problem**, which may affect model training and evaluation.

2. Missing Value Analysis

The dataset contains several missing values that should be addressed before model development.

Variable	Missing Values	Risk Level
Income	39	High
Loan_Balance	29	Medium
Credit_Score	2	Low

Recommended Treatment

- **Income:** Impute using median values grouped by employment status or credit profile.
- **Loan_Balance:** Median imputation or predictive imputation.
- **Credit_Score:** Mean/median imputation due to very low missing percentage.

Potential Data Quality Concerns

- Employment status values contain inconsistent labels such as:
- EMP

- `Employed`
- `employed`
- These should be standardized before modeling.

3. Key Patterns Observed

Financial Risk Trends

1. Customers with higher debt burdens and utilization levels show slightly elevated delinquency risk.
2. Business and student credit card holders demonstrate higher delinquency rates compared with platinum card holders.
3. Unemployed customers show higher delinquency incidence than employed groups.
4. Recent repayment behavior (`Late` and `Missed`) appears useful for predicting future delinquency.

Repayment Behavior Insights

- Monthly repayment variables (`Month_1` to `Month_6`) contain repeated behavioral patterns that are likely valuable predictors.
- Customers with repeated `Late` or `Missed` payments tend to show higher delinquency likelihood.

4. Outliers and Anomalies

Potential Outliers

- Extremely high income values compared with the overall population.
- Large loan balances relative to customer income.
- High debt-to-income ratios exceeding expected lending thresholds.

Data Anomalies

- Inconsistent employment status categories (`EMP`, `employed`, `Employed`).
- Possible noisy repayment behavior where some customers marked `On-time` still appear delinquent.
- Mild inconsistencies between missed payments and delinquency outcomes.

These issues may reduce model interpretability and predictive stability if not cleaned.

5. Fields That May Cause Problems for Delinquency Modeling

Field	Potential Issue
Customer_ID	Unique identifier with no predictive value; should be removed

Field	Potential Issue
Employment_Status	Inconsistent category naming may create duplicate classes
Location	May introduce bias or weak generalization if overfitted
Month_1 to Month_6	High dimensional categorical behavior may require encoding
Income	Missing values could bias predictions
Class imbalance in Delinquent_Account	Model may favor non-delinquent predictions

Additional Modeling Risks

- Target imbalance may reduce recall for delinquent customers.
- Categorical variables require encoding before machine learning.
- Missing financial data may distort risk patterns.
- Sequential month variables may create multicollinearity.

6. Top 3 Variables Most Effective for Predicting Delinquency

Based on behavioral relevance, observed patterns, and predictive usefulness:

1. Monthly Repayment History (Month_1 to Month_6)

- Strong behavioral indicator of repayment consistency.
- Repeated late or missed payments are highly associated with future delinquency.
- Likely the strongest feature group for prediction.

2. Credit_Utilization

- High utilization often reflects financial stress.
- Frequently used in credit risk scoring models.
- Useful for identifying overleveraged customers.

3. Debt_to_Income_Ratio

- Measures repayment burden relative to income.
- High values indicate increased financial pressure and repayment risk.

Other Potentially Useful Variables

- Missed_Payments
- Employment_Status
- Credit_Score
- Loan_Balance

7. Recommended Next Steps

1. Clean and standardize categorical fields.
 2. Handle missing values using robust imputation methods.
 3. Address class imbalance using:
 4. SMOTE
 5. Class weighting
 6. Balanced sampling
 7. Encode categorical variables.
 8. Engineer repayment trend features from monthly status variables.
 9. Evaluate models using:
 10. AUC-ROC
 11. F1-score
 12. Recall
 13. Use explainability techniques such as SHAP to validate feature importance.
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8. Reference Notes

This summary follows the structure outlined in the EDA report template and glossary documents provided by the user.